



Portable device tests for gluten

Using the Nima device, individuals can make sure that their food is gluten free, by analysing a tiny sample.
http://dgit.in/gluten_freex



Cars need antivirus too

Connected cars are not only susceptible to hacking, but also to viruses.
http://dgit.in/connected_carsx

The difference suggests the presence of dark matter. Because of the high amount of sharpness and detail a JWST image can produce, it becomes a good candidate to make lensing measurements. Scientists also use high fidelity simulations to study the role of dark matter in the evolution of the universe. Data from galaxy formation in the early universe, something that the JWST is primed to do, can test these predictions and lead to better models. The JWST will be able to observe supernovae that are very dim and thus supplement other data that ties the red shift of supernovae to the existence of dark energy. While the JWST can perform most of the functions that the HST currently performs, it also stretches the ability of a telescope by a fair amount and allows scientists a view of the cosmos like never before.



Captured image from the HST (top panel) of an interstellar disc that is seen during star and planet formation. Artists impression (Bottom Panel). JWST will give us better data from these systems.

Once the JWST is successfully launched in space we possess the ability to reverse the clock momentarily and study how the universe as we know it came to be, even if only in dull infrared.

Who made the JWST and who gets to use it?

A project like the JWST is far from being a one team, or even a one nation job. It is a successful collaboration of a large consortium of scientists, engineers and managers from different countries. NASA, ESA

(European Space Agency), The Canadian Space Agency and a host of other private companies are responsible for assembling thousands of sophisticated instruments into what can only be described as the most advanced instrument built by man to study space. Officially fourteen countries have contributed in one way or another to the JWST, though all countries are from either North American or Europe. Even though the making of the JWST was not truly global, in the fullest sense of the word, the telescope is a global observatory and will allow scientists from around the world to submit proposals to use the JWST. A peer review system will rank the proposals according to priority, much like the current system adopted for using the HST. Major scientific discoveries will be announced at Press releases, research studies will be published in journals and data will be made available on the internet for research groups across the world to use. Also, like the HST frequently released stunning images, the JWST will do the same.

How long will the JWST operate for and what if it fails?

In its current condition the JWST is built to carry enough fuel to keep it running till 2028. Assuming it makes a successful scheduled launch in 2018, the JWST could provide scientists with the most fruitful decade in the history of cosmology and astronomy.

There is no denying the fact however that a complicated machine like JWST could fail in so many ways. The mirror could degrade, compromising the light sensing ability of the telescope. Even minute failure of the sunshield or leakage of the coolant can critically damage the telescopes IR sensing functionality. Then there is the looming danger of a problem during deployment or launch. The people behind the JWST however, have completed extensive testing of every piece of equipment. From simulation tests to actual experiments, all possible methods have been used to expose the telescope to the harshest of tests. But what if the telescope fails in space?

Because of the distance from Earth at which the JWST will be parked, it is not

possible (with our current technology) to launch a repair mission for the telescope. We cannot however, discount the possibility of robotic technology making the required leaps in the near future. The truth is, a majority of space missions this far have been launched into space without the existence of an appropriate repair mechanism. Thanks to the professionalism of the team behind the JWST, its chances of success are looking good, and hopefully the JWST will launch successfully.

How does the JWST compare with the Thirty Metre Telescope (TMT)?

The TMT when completed will be the worlds second largest telescope (the largest being the E-ELT in Chile (still under construction)) having a primary mirror that



The Pillars of Creation : One of HST's most famous images showing interstellar gas forming in the Eagle Nebula

is much larger than the one on-board the JWST. The TMT also has excellent range and can detect light from near UV to mid IR. It is however an Earth based telescope which makes it unable to completely get rid of the effects of atmospheric dust and pollution. The JWST being space based, doesn't have such problems, also installing a host of spectral measurement instruments. The TMT, which has since its inception shifted base to Ladakh, is excellent news for astronomers with its enhanced light capturing facility allowing scientists to probe deeper into the universe than any Earth telescope before it. **d**